

**Table 15 – Test requirements for mechanical characteristics
of insulating compounds (before and after ageing)**

Designation of compounds (see 4.2)	Unit	PVC/A	EPR		HEPR		XLPE	
			0,6/1(1,2) kV cables with copper conductors	All other cables	0,6/1(1,2) kV cables with copper conductors	All other cables	0,6/1(1,2) kV cables with copper conductors	All other cables
Maximum conductor temperature in normal operation (see 4.2)	°C	70	90	90	90	90	90	90
<i>Without ageing</i> (IEC 60811-501)								
Tensile strength, minimum	N/mm ²	12,5	4,2	4,2	8,5	8,5	12,5	12,5
Elongation-at-break, minimum	%	150	200	200	200	200	200	200
<i>After ageing in an air oven</i> (IEC 60811-401)								
After ageing without conductor								
Treatment:								
– temperature	°C	100	135	135	135	135	135	135
– tolerance	K	±2	±3	±3	±3	±3	±3	±3
– duration	h	168	168	168	168	168	168	168
Tensile strength								
a) value after ageing, minimum	N/mm ²	12,5	–	–	–	–	–	–
b) variation ^a , maximum	%	±25	±30	±30	±30	±30	±25	±25
Elongation-at-break:								
a) value after ageing, minimum	%	150	–	–	–	–	–	–
b) variation ^a , maximum	%	±25	±30	±30	±30	±30	±25	±25
After ageing with copper conductor followed by the tensile test ^b				–				
Treatment:								
– temperature	°C	–	150	–	150	–	150	–
– tolerance	K	–	±3	–	±3	–	±3	–
– duration	h	–	168	–	168	–	168	–
Tensile strength:								
Variation ^a , maximum	%	–	±30	–	±30	–	±30	–
Elongation-at-break:								
Variation ^a , maximum	%	–	±30	–	±30	–	±30	–
After ageing with copper conductor followed by bending test (only if the tensile test is not practicable) ^b								
Treatment:								
– temperature	°C	–	150	–	150	–	150	–
– tolerance	K	–	±3	–	±3	–	±3	–
– duration	h	–	240	–	240	–	240	–
Results to be obtained		–	No cracks	–	No cracks	–	No cracks	–
^a Variation: difference between the median value obtained after ageing and the median value obtained without ageing expressed as a percentage of the latter.								
^b See 18.4.2.								

Table 18 – Test requirements for mechanical characteristics of sheathing compounds (before and after ageing)

Designation of compound (see 4.3)	Unit	ST ₁	ST ₂	ST ₃	ST ₇	ST ₈	SE ₁
Maximum conductor temperature in normal operation (see 4.3)	°C	80	90	80	90	90	85
<i>Without ageing</i> (IEC 60811-501)							
Tensile strength, minimum	N/mm ²	12,5	12,5	10,0	12,5	9,0	10,0
Elongation-at-break, minimum	%	150	150	300	300	125	300
<i>After ageing in an air oven</i> (IEC 60811-401)							
Treatment:							
– temperature (tolerance ±2 K)	°C	100	100	100	110	100	100
– duration	h	168	168	240	240	168	168
Tensile strength:							
a) value after ageing, minimum	N/mm ²	12,5	12,5	–	–	9,0	–
b) variation ^a , maximum	%	±25	±25	–	–	±40	±30
Elongation-at-break:							
a) value after ageing, minimum	%	150	150	300	300	100	250
b) variation ^a , maximum	%	±25	±25	–	–	±40	±40
^a Variation: difference between the median value obtained after treatment and the median value without treatment, expressed as a percentage of the latter.							